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I have a problem with my account, which has been disabled, and I was in contact with Rachel. Unfortunately, nothing has happened yet.

I was able to follow up with you through the sessions and labs until the load balancer lab, but nothing after that. So, I hope I can describe the rest of the points in text.

I've taken 3 screenshots from my account that may be useful.

Q 1/

I logged into my account and created a resource group named **web app**. This resource group will be the source for the web app and other stuff that I'm creating. After that, I clicked the + sign to create a web app, and I could choose from many of the VMs. I went through all the steps, like selecting the resource group, code type, region, and jumping through the network and monitor functions until creating the app. I started the app and then went to the deployment section to get the credentials for this app.

Then, I used a program called **FileZilla** to connect to the web app by using the credentials. I used **Visual Studio Code** to get a code representing the web page to show some information about the page. In **FileZilla**, the left side represents the localhost, and the right side represents the remote server. I just moved the code file I made to the **wwwroot** side, and it automatically uploaded to Azure. When you click the link again, you will see the new version of the website with the changes you made.

Name	Type	Last Viewed
 WebApp	Resource group	3 minutes ago

Resources		Recommendations	
Filter for any field...		Type equals all X	Location equals all X + Add filter
Showing 1 to 4 of 4 records. ☐ Show hidden types ⓘ		No grouping ▼	List view ▼
☐ Name ↑↓	Type ↑↓	Location ↑↓	
☐  Application Insights Smart Detection	Action group	Global	...
☐  ASP-WebApp-b20a	App Service plan	North Europe	...
☐  webapp001	App Service	North Europe	...
☐  webapp001	Application Insights	North Europe	...

Q 2/

I couldn't view my work due to the disabled account issue. But I'll put 2 screens shot

Resources

Recent Favorite

Name	Type	Last Viewed
 NetworkWatcher_northeurope	Network Watcher	a minute ago
 NetworkWatcherRG	Resource group	4 hours ago
 webapp001	App Service	4 hours ago
 WebApp	Resource group	4 hours ago
 RGEc	Resource group	4 hours ago
 Azure for Students	Subscription	2 weeks ago
 rgec02	Virtual machine	2 weeks ago
 rgec-vnet	Virtual network	2 weeks ago
 rgec	Virtual machine	2 weeks ago
 RG-VM	Resource group	2 weeks ago
 rgecvnet1	Virtual network	2 weeks ago

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 **NetworkWatcher_northeurope** ☆ ☆ ...

Network Watcher

×

^ Essentials

[JSON View](#)

 Overview

 Activity log

 Access control (IAM)

 Tags

> Settings

> Monitoring

> Automation

> Help

Resource group [\(move\)](#) : [NetworkWatcherRG](#)
Location [\(move\)](#) : North Europe
Subscription [\(move\)](#) : [Azure for Students](#)
Subscription ID : 82659b7c-c76c-4674-8c1b-51f2a1e4ba62
Tags [\(edit\)](#) : [Add tags](#)

Q 3/

A /

RTO is a shortcut for **Recovery Time Objective**, which means the time that can be offered by the Azure service owner for the service to be down. The shorter the time you plan, the better the availability for your services. This time depends on several factors

like network bandwidth, storage size, and location. To meet your desired RTO, it may be necessary to optimize the backup schedule and use data replication across multiple regions to get it quicker and then use faster storage. Moreover, practicing data backup will help achieve the desired time objectives in our case.

B /

RPO is a shortcut for **Recovery Point Objective**. In other words, it refers to the amount of data that can be tolerated for loss by the service owner. For example, if the time is fixed to 15 minutes, it means that every 15 minutes the data will be backed up, and when the service is down for any reason, the maximum data loss will be 15 minutes.

There are two types of RPO: **frequent backups** and **infrequent backups**. When we refer to **frequent backups**, it means that we set up a backup time every 15 minutes, which leads to a lower RPO. When we set the backup time to once a day, this leads to a higher RPO. A **lower RPO** means less data will be lost when the service is down.

C /

Retention means the period that your data will be available until it is deleted by the service owner in Azure. There are many types of retention in Azure, such as short-time retention, long-time retention, or infinite retention. When we're talking about short-time retention, it means a weekly, daily, or monthly period. This type of retention considers information to be less important to keep for a long time. With long-time retention, we are talking about months or years, depending on the information's sensitivity, such as information for bank users and transactions. Lastly, infinite retention would be used for special cases, like databases or VM backups.

Setting up a backup policy, and especially retention, relies on many factors, and one of them is the storage cost. This means that the longer the retention, the more data storage it will cost. Lastly, the retention policy should be considered based on business needs and data protection.

D /

In **Network Watcher**, under the diagnostic tools, we can choose **IP Flow**, which is a type of service in Azure that allows us to check if a packet is passed or denied from the virtual machine based on five-tuple information. If the packet is denied, we will automatically get the security group decision and the name of the rule as return information. We can specify the virtual machine, the network interface, and then choose

the type of protocol that we want to test, along with the direction (inside or outside the virtual machine). Then, we can select the local port and finally verify the IP flow to see if we have received the packet or not.

Meanwhile, we can find the **Connection Troubleshoot** under **Network Watcher**, which is a service provided by Azure to show a direct TCP or ICMP connection from a virtual machine. We can go through some steps, like selecting the source type and the virtual machine, then selecting the destination type and the virtual machine. In simple terms, it's the destination and source target between both virtual machines. Then, we can choose both IP versions or one of them, followed by selecting the type of protocol (TCP or ICMP, which is used for ping). Finally, we can select the destination and source port, and after that, we can perform a diagnostic test to get the results.

Both **IP Flow** and **Connection Troubleshoot** aim to show connectivity, but if we are seeking accurate use, as I understand it, **IP Flow** will show if there are any rules, like blocking or restricted areas, between the source and destination targets, while **Connection Troubleshoot** shows if we have a connection between the source and destination targets.

Unfortunately, I cannot provide screenshots due to the same reason I mentioned at the beginning of the document.

E /

Both **crash consistent** and **application consistent** are terms describing a data backup in Azure. These two terms are critical to ensure that the data in Azure will be backed up smoothly and accurately. We can go over some points regarding **crash consistent** and **application consistent** backup:

1. **The speed of backup:** It should be faster in **crash consistent** because we don't need the application status, and slower in **application consistent** because we need the application status.
2. **The recovery process:** **Crash consistent** may need some further action and process after the backup is completed, while in **Application consistent**, no additional process or action is needed.
3. **The data consistency:** The most important thing for **crash consistent** is that the operating system data is in a consistent state, with no guarantee for application data, while ensuring both the operating system and the application data is critical in the **application consistent** backup.
4. **Use case:** **Crash consistent** is suitable for non-critical data, while **application consistent** is suitable for critical services that require exact data consistency.

F /

Both **Web App Service** and **IIS (Internet Information Services)** represent two methods of hosting and managing web applications. While **Web App Service** represents a **Platform as a Service**, **IIS** represents **Infrastructure as a Service**. I think the main difference between the two services is that in **IIS**, we have full control over the settings and the server environment, while in **Web App Service**, we leave some customization for Azure to manage.

I will list some points to describe the difference between the **Web App Service** and **IIS** regarding what we need to do in the VM:

1. **Software installation:** In **IIS**, we need to understand and install the required software manually, while in **Web App Service**, Azure handles the software installation.
2. **Security:** In **IIS**, we are responsible for applying the security configurations, while in **Web App Service**, Azure handles the security configurations and updates.
3. **Backup:** In **IIS**, we manage the backup for the VM manually, while in **Web App Service**, Azure provides a backup feature.
4. **Deployment:** In **IIS**, we deploy manually, while in **Web App Service**, we can deploy via **GIT**, **CI/CD**, or **Visual Studio**.
5. **Monitoring:** In **IIS**, we set up the monitoring tools manually, like **IIS logs**, while in **Web App Service**, built-in tools are provided by Azure.
6. **Scaling:** In **IIS**, if scaling is needed, we must resize the VM manually to adjust resources, while in **Web App Service**, we can use the auto-scaling rules provided by Azure with automatic load balancing.